

Category, Not Severity: How Large Language Models Misclassify AI Incidents Under the EU AI Act Risk Taxonomy

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Abstract

The EU AI Act assigns risk categories by what an AI system does and where it is deployed, not by how much harm it has caused. Incident trackers increasingly assign these categories with large language models (LLMs), yet a recent audit found several frontier LLMs scoring below chance on the Act’s category while doing well on other taxonomies. To test whether the failure is procedural, a lawyer labelled 144 incidents from the AI Incident Database against a published rubric derived from the Act, and 14 LLMs classified them under a basic prompt and under a step-by-step prompt that walks the Act’s rules in order. Agreement with the expert ranges from $\kappa = -0.07$ to 0.65 under the basic prompt and from 0.30 to 0.70 under the step-by-step prompt (means 0.42 and 0.50). We show that the errors are directional: LLMs over-classify incidents outside Annex III and stay unbiased inside it, as expected if they judge severity rather than category. Spelling the rules out roughly halves that outward bias, at the cost of a smaller opposite one inside Annex III. For automated monitoring, categories assigned outside Annex III are the first place to spend human review. Rubric, prompts and labels are released.

Keywords

EU AI Act, risk classification, large language models, AI incidents, regulatory compliance, AI governance

1 Introduction

AI incident databases have become policy infrastructure. The AI Incident Database (AIID) collects public reports of cases in which a deployed AI system caused or nearly caused harm (**incidents**), each stored as a short structured record linked to the underlying news coverage [7]. Trackers built on top of it, such as the MIT AI Incident Tracker [8], attach classification schemes (**taxonomies**) to every incident, and downstream readers rely on the resulting labels. One of those taxonomies is the risk category of the EU AI Act [4] — hereafter the Act; “Art. 6(3)” means paragraph 3 of its Article 6 — which sorts AI systems into four categories: *prohibited*, *high-risk*, *transparency* and *minimal*, as seen in Figure 1. Annotating thousands of records by hand is infeasible, so these labels are increasingly produced by large language models (LLMs).

This shortcut has a documented weak spot. In June 2026 the MIT AI Risk Initiative audited its own tracker against two human reviewers and reported that four of eight frontier LLMs scored a *negative* quadratic weighted kappa — worse than chance — on

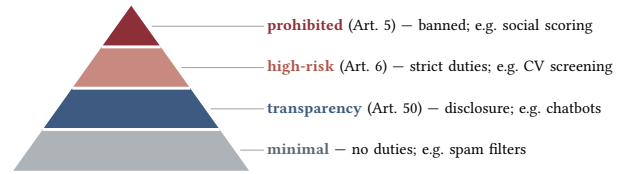


Figure 1: The four risk categories of the EU AI Act, with typical systems. How severe an incident’s real-world outcome was plays no part in the assignment.

the AI Act category while clearing the human baseline on other taxonomies [8]. The audit rested on ten incidents, on which the prompt refinements that repaired the score were also developed — a limitation the authors state while calling for held-out data.

We take up that invitation and add a hypothesis about *why* the failure occurs. The Act’s classification logic is categorical and function-based. A system is high-risk because it is a safety component of a product that already undergoes third-party safety certification, such as machinery or a medical device (Art. 6(1), **Annex I**), or because it is used in one of eight sensitive application areas — among them employment, education, credit and law enforcement (Art. 6(2), **Annex III**) — subject to a narrow exception for purely preparatory or procedural roles (Art. 6(3)), unavailable where the system profiles people. How severe the real-world consequences of an incident were is not a factor in this classification.

We hypothesise that LLMs nevertheless tend to classify incidents by the severity of the outcome — the question a short record lets them answer — rather than by the category question the statute asks. We call this **severity substitution**. If it occurs, the errors should be systematic rather than random: over-classification of severe-sounding incidents outside Annex III, and under-classification of mundane-sounding incidents inside it.

Our contributions are (i) a lawyer-written, published annotation rubric that turns the Act’s classification rules into a procedure for short incident records; (ii) an expert-labelled sample of 144 AIID incidents, fourteen times the size of the audit above; (iii) a 14-LLM comparison under a basic and a step-by-step prompt; and (iv) an error analysis by Annex III membership that locates where severity substitution occurs.

2 Related Work

Annotating AI incidents. The AIID was built to make AI incidents citable and comparable [7], and the taxonomies attached to its records now feed risk repositories and policy reporting [11]. Most such taxonomies describe what happened: the MIT AI Risk Repository’s domain taxonomy, which the tracker of Section 1 applies alongside the Act’s category, sorts a record into one of seven domains such as discrimination, privacy or misinformation [11]. The Act’s category is different in kind: it asks what the

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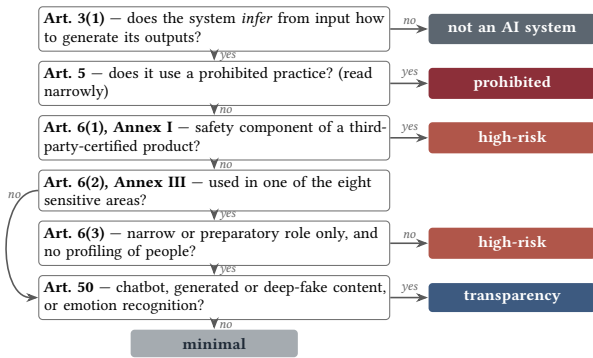


Figure 2: The classification procedure that the rubric encodes and the step-by-step prompt follows. Art. 50 duties are recorded even where a higher category already applies.

system is for, not what it did. Paeth et al. [10] review more than 750 incident records under two descriptive taxonomies and find that uncertainty about cause, extent of harm and technical detail is unavoidable in incident reporting. That caps how far any two annotators — human or machine — can agree, and makes “the record does not say” a substantive answer rather than a failure.

Automating AI Act classification. The closest prior work is CORTEX [9], a taxonomy of 29 technical vulnerability groups applied to over 1,200 AIID incidents, each mapped onto the AI Act and related frameworks and given a composite *risk score* that is explicitly severity- and exposure-weighted, not validated against legal judgements of the Act’s own categories. Our question is the complement of theirs: not how risky a system is, but which category the statute assigns it to, a question in which severity plays no part.

The difficulty of the classification. Leinarte [6] shows how much interpretive work the high-risk rules (Art. 6 and Annex III, Section 3) demand in practice. Empirically, experts applying a pre-final draft of the Act could not clearly classify roughly 40% of 106 real corporate AI systems [1], and the MIT audit reports a human-human agreement of only $\kappa = 0.56$ on AI Act categories [8]. The task is genuinely hard for people, which is why we frame results as expert-model disagreement rather than model accuracy. A structured, question-by-question assessment has been proposed for related governance judgements [3]; our step-by-step prompt tests whether the same structure helps LLMs with the Act.

3 The Classification Rules and the Annotation Rubric

The rubric answers a hypothetical question: if the system involved in this incident were placed on the EU market today, which risk category would it fall under? The Act is assumed to apply — most incidents predate it or occurred outside the EU — and the object of classification is the system and its intended purpose, not the harm in the record. Figure 2 sketches the procedure; below, how each step is read on incident records.

Is it an AI system at all? A purely deterministic, rule-based process is not an AI system under Art. 3(1) and is recorded as such; the definition is broad, so this disposes of few incidents.

Does it use a prohibited practice? The eight prohibited practices of Art. 5 — social scoring and untargeted scraping of facial images among them; not to be confused with the eight sensitive areas of Annex III — are read narrowly, as prohibitions

must be. Under Art. 5(1)(a), a subliminal or purposefully manipulative technique must materially distort behaviour by *appreciably* impairing informed decision-making and thereby cause, or be likely to cause, *significant* harm. Intent to harm is not required, but the technique must be attributable to how the system was trained or configured; ordinary hallucination does not qualify. A third party who misuses a general-purpose generator to run a fraud is therefore a *user* of a tool, not the provider of a prohibited system — such incidents fall under Art. 50 instead; a system *configured* for the deceptive purpose remains prohibited.

Is it a safety component of a certified product? High-risk status under Art. 6(1) attaches to AI embedded in product types that must pass third-party safety certification before sale (Annex I), such as medical devices, lifts, toys and vehicles. Incident records rarely say whether a product was certified, so the rubric asks what the product type normally requires: a car’s automated-driving system qualifies, ordinary industrial machinery does not, because its maker certifies it without a third party.

Is it used in a sensitive area, and does the exception apply? Membership of the eight sensitive areas (employment, education, credit, migration and law enforcement among them) is judged by intended purpose, not by who happens to hold the system: criminal misuse of a tool built for law enforcement does not place the incident in the law-enforcement area, and where two areas overlap, the system’s underlying purpose decides. Content moderation and product recommendation belong to the Digital Services Act and consumer law, outside Annex III. On the Art. 6(3) exception, flagging people for human investigation is normally *not* exempt: a system that selects who is flagged influences the assessment, and such systems generally profile people, which closes the exception; silence in the record does not open it.

Are transparency duties owed? Whether the system is one Art. 50 covers — a chatbot, a generator of synthetic or deep-fake content, an emotion-recognition system — is recorded independently of the category, since a system may owe transparency duties and be high-risk at the same time: a recruitment tool with a chatbot front end is both. The reported category is the higher one; the Art. 50 answer survives in the per-step labels. The rubric records finer points than fit here; the full document, on which both the labels and the step-by-step prompt are based, is published with the data and code.¹

4 Data and Method

Sample. We work from the AIID’s official spreadsheet export of 10 August 2026. From it we drew 150 incidents at random under a fixed seed, stratified by simple keyword matching on titles and descriptions (“hiring” suggests employment, “police” law enforcement, and so on) so that all eight sensitive areas and a control group outside them would be represented. These keyword buckets are only a sampling device — often wrong about the true area — and were hidden from the annotator. An incident record carries a title, a one-paragraph description (median 358 characters) and the parties involved; its full AIID page adds the linked news coverage.

Rubric development. The rubric was developed on six of the 150 incidents, its rules drafted, tried against them and refined until stable. Because the annotator’s reading of those six formed during drafting, they are excluded from evaluation and published as worked examples; the remaining 144 form the evaluation set, labelled under a rubric that was not changed again.

¹<https://github.com/stefankofilovski-lang/category-not-severity>

Incident S058 (AIID #599, 2023). “An industrial robot is reported to have crushed a man to death in South Korea when it failed to differentiate the man from the boxes of produce it was handling.”

Expert (per rubric): no prohibited practice; not an Annex I product — stacking robots normally self-certify, and nothing shows a self-learning safety component needing third-party certification; no sensitive area; no transparency duty ⇒ **minimal**, confidence 3/3.

LLMs: basic prompt — 12 of 14 answer **high-risk** (two: *not an AI system*); step-by-step prompt — 10 of 14 answer **minimal** (four: high-risk).

Figure 3: A worked example: maximal severity, minimal category.

Basic prompt (abridged)

Classify each of the following AI incidents according to the European Union AI Act risk level taxonomy. [...] assign exactly one risk level: “prohibited” [...] “minimal” [...]

Step-by-step prompt (the steps of a ~2,100-token instruction)

GATE – Art. 3(1). Is this an AI system at all [...]

STEP 1 – Art. 5(1): prohibited practices [...]

STEP 2 – Art. 6(1)/Annex I: [...] third-party conformity assessment? [...]

STEP 3 – Art. 6(2)/Annex III: which of the eight areas, if any [...]

STEP 4 – Art. 6(3) derogation. [...] **STEP 5** – profiling override [...]

STEP 6 – Art. 50 transparency. [...]

STEP 7 – otherwise → risk_tier “minimal”.

Figure 4: The two prompts, abridged (full text in the repository).

Expert labels. One annotator, a lawyer (the first author), labelled all 144 incidents, recording an answer to each of the six questions of Figure 2, followed by the final category, his confidence on a three-point scale, and whether the description sufficed or the full AIID page had to be consulted (needed for 14 of the 144). The annotator and the LLMs alike could also answer *not an AI system* or *insufficient information*, since incident reports are often silent on facts the Act needs. The resulting class distribution is 77 transparency, 44 high-risk, 10 prohibited, 9 minimal, 2 *not an AI system* and 2 *insufficient information*. After labelling, a script checked each record for internal consistency with Figure 2 — a record answering *yes* to the prohibited-practice question, for instance, must end in the *prohibited* category — and a review compared records with materially identical facts. Together they found 19 fields on 10 incidents that broke one of these rules; each was corrected, and every correction is published with its reason. We do not present these labels as ground truth: what we measure is where LLMs disagree with one expert reading of the Act, a limitation taken up in Section 6.

Two prompts. Every incident was classified by every LLM under two fixed prompts that share the same answer options and differ only in whether the legal procedure is spelled out (Figure 4): the *basic prompt* is a minimal instruction of the kind the audited tracker itself uses [8]; the *step-by-step prompt* walks the procedure of Figure 2, requiring an answer at every step before the category. Both predate all labels and are published verbatim with the rubric, labels, corrections and code.

Models and measures. We evaluated 14 LLMs through the OpenRouter API — plain completions, with no tools or web access — including the six from the MIT audit (Section 1) for direct comparison. Every model classified every incident under both prompts, with one exception: Gemini 3 Flash declined two incidents on content-policy grounds under the basic prompt, on repeated attempts; its scores there are computed over the 142 it answered. Agreement between an LLM and the expert is quadratic weighted kappa (κ) [2] on the ordered scale minimal < transparency < high < prohibited (1 perfect agreement, 0 chance

Table 1: Agreement between each LLM and the expert (quadratic weighted kappa) on all 144 incidents, under both prompts. †: the six models from the MIT audit [8]; *: declined two incidents under the basic prompt (Section 4).

Model	Basic	Step-by-step
Gemini 3.7 Flash	0.65	0.70
Grok 4.6	0.47	0.67
Kimi K2.5†	0.45	0.65
Qwen 3.8 Max	0.39	0.58
Qwen 3.8 27B	0.47	0.58
Gemini 3 Flash†*	0.57	0.53
Claude Haiku 4.5†	−0.07	0.49
Claude Sonnet 5	0.32	0.48
GPT-5.2†	0.57	0.48
Kimi K3	0.54	0.42
Claude Sonnet 4.6†	0.24	0.40
GPT-5.6 Sol	0.48	0.37
Gemma 3 27B†	0.28	0.31
GPT-5.6 Luna	0.56	0.30

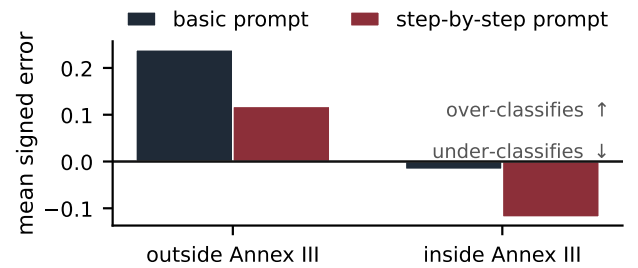


Figure 5: Mean signed category error (LLM minus expert), outside versus inside Annex III.

level, negative worse than chance; a two-category miss costs more than a one-category one), chosen because transparency alone holds 53% of the labels and raw accuracy would reward always answering it. We add percentile bootstrap intervals and exact-match shares (the percentage of answers identical to the expert’s). *Not an AI system* and *insufficient information* have no rank on this scale — ranking them would invent an order the Act does not define — so those pairs are reported as coverage, not scored; they are rare (3% of basic-prompt answers, 1% step-by-step). The error-direction analysis takes the signed difference between LLM and expert category (+1 = one too high), split by Annex III membership as judged by the expert.

5 Results

Agreement. Table 1 reports each LLM’s agreement with the expert under both prompts. Under the basic prompt the spread is wide, from −0.07 to 0.65; under the step-by-step prompt it narrows to 0.30–0.70. Against the human–human baseline of 0.56 — moderate agreement on the conventional scale [5] — the best LLMs under either prompt sit at or slightly above what two human reviewers achieve.

Error direction under the basic prompt. Figure 5 shows the central result. (Annex III lists the eight sensitive areas; an incident is *inside* it when the expert placed its system in one of them.) Under the basic prompt:

- *outside* Annex III, 20.2% of answers land one or more categories too high and 8.9% too low — a mean signed error of +0.24 categories;
- *inside* Annex III, answers are balanced: 7.6% too high, 7.4% too low, mean −0.02.

The asymmetry is not explained by difficulty: exact-match agreement is 85.0% inside Annex III against 70.9% outside it. LLMs are accurate where the statute supplies a category: agreement is perfect on the essential-services and education incidents and reaches 96% on employment. Incident age tracks the same split: exact-match agreement is 82% on pre-2020 incidents but 70% from 2023 onwards, where generative-AI misuse outside Annex III dominates.

What basic-prompt errors look like. Fraud incidents — an AI-driven romance-scam campaign in Arizona, a HK\$34M deep-fake fraud in Hong Kong, a phishing operation impersonating Australian government services — are classified as *prohibited* by 10, 9 and 9 of the 14 LLMs under the basic prompt, two categories above the *transparency* answer the rubric gives: the criminal is a user of an off-the-shelf tool, not the provider of a prohibited system (Section 3). The fatal robot accident of Figure 3 is misread the same way.

Step-by-step results. Spelling the rules out removes much of the outward drift: outside Annex III the mean signed error falls from +0.24 to +0.12, over-classification from 20.2% to 13.1%, and exact-match agreement rises from 70.9% to 80.7%. Mean kappa across the panel rises from 0.42 to 0.50, and the gain is concentrated where the basic prompt was weakest: the model that scored below chance reaches $\kappa = 0.49$, and every LLM in the bottom half of the basic-prompt ranking improves. Agreement improves for 9 of the 14 overall. The three frauds above are still called *prohibited* by 7, 8 and 3 of the 14.

Step-by-step errors. The step-by-step prompt's own errors sit inside Annex III, on incidents that sound routine: for a UK passport-application photo checker that rejected dark-skinned applicants' photos (high-risk, migration area), LLMs apply the Art. 6(3) exception more readily than the rubric allows and land too low. Inside Annex III the step-by-step prompt under-classifies 12.6% of the time against the basic prompt's 7.4%, turning a mean error of -0.02 into -0.12 . The correction is thus neither free nor uniform. Five LLMs lose ground under the procedure — all three GPT models, Kimi K3 and Gemini 3 Flash, the largest drop 0.27 — while both Claude and both Qwen models, Grok, Kimi K2.5 and Gemini 3.7 Flash gain. All five were top-half models under the basic prompt, but the split follows the provider more than the rank: the walk-through supplies structure some models lacked and constrains others that were right without it.

6 Discussion and Limitations

Our findings show the limit of basic prompting and support the severity-substitution hypothesis at the level of the panel: asked for an AI Act category with no procedure attached, LLMs on average behave as if asked how severe the harm was. Where the Act supplies a category this does little damage, because Annex III incidents tend to be serious as well as covered; where it supplies none — the frauds and the robot accident of Section 5 — severity has nothing to anchor to, and the model reaches for a higher category. This is the costly direction of error for anyone publishing model-assigned categories: it inflates the apparent prevalence of prohibited and high-risk systems in exactly the records that attract attention. We also partially replicate the MIT audit [8]: one panel model falls below chance on our data — not one of their four — while two of their below-chance models are among our strongest.

Several limitations bear on how far this generalises. The gain is not uniform: all three GPT models do worse with the procedure (Section 5), so the panel average recommends no particular model, and why providers differ needs a larger panel to answer. The labels come from a single annotator; publishing the rubric, the per-step records, the confidence ratings and the 19 corrected values makes every judgement checkable — not replicated — and the annotator's confidence is lowest on the contested records. One classification — third-party misuse of general-purpose generative systems under Art. 50, not Art. 5 — is the first author's interpretive choice rather than settled law, and reversing it inverts the model ranking; the per-step records let a reader of the opposite view re-score the data without new annotation.

Two further limits come from the data. Fifteen of the 44 high-risk incidents are automated-driving cases, high-risk through the certified-product route, so agreement on that class partly reflects one recurring pattern rather than 44 independent judgements. And incident reports are often silent on the facts the Act needs (Section 4), which caps what any classifier can recover.

7 Conclusion and Future Work

Frontier LLMs do not fail at EU AI Act classification uniformly but directionally: they over-classify incidents outside the Act's sensitive areas and stay calibrated inside them. An explicit procedure removes most of that drift at the price of a smaller opposite bias. The practical rule for anyone attaching AI Act categories with an LLM: spend the human review outside Annex III first.

Three directions for future work follow. The step-by-step prompt's new inward errors concentrate in the Art. 6(3) exception, so a prompt demanding record evidence before granting it is the obvious next iteration; a non-LLM baseline would show how much agreement surface features alone achieve; and where the Act is genuinely open to interpretation, a classifier should perhaps return the range of defensible categories rather than one answer. The per-step records released here are raw material for all three.

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References

- [1] appliedAI Initiative. 2023. AI Act Risk Classification Study. Tech. rep. <https://www.appliedai.de/uploads/files/AI-Act-Risk-Classification-Study-EN.pdf>.
- [2] Jacob Cohen. 1968. Weighted kappa: nominal scale agreement provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70, 4, 213–220. doi:10.1037/h0026256.
- [3] Marios Constantinides et al. 2024. Good intentions, risky inventions: a method for assessing the risks and benefits of AI in mobile and wearable uses. *Proc. ACM Hum.-Comput. Interact.*, 8, MHCI, 1–28. doi:10.1145/3676507.
- [4] European Parliament and Council of the European Union. 2024. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). OJ L, 2024/1689. (2024). <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>.
- [5] J. Richard Landis and Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics*, 33, 1, 159–174. doi:10.2307/2529310.
- [6] E. Leinarte. 2024. The classification of high-risk AI systems under the EU Artificial Intelligence Act. *J. AI Law and Regulation*, 1, 3, 262–280. doi:10.21552/aire/2024/3/4.
- [7] Sean McGregor. 2021. Preventing repeated real world AI failures by cataloging incidents: the AI incident database. In *Proc. AAAI Conf. on Artificial Intelligence*. Vol. 35, 15458–15463. doi:10.1609/aaai.v35i17.17817.
- [8] MIT AI Risk Initiative. 2026. AI incident tracker: June 2026 update. <https://airisk.mit.edu/blog/ai-incident-tracker-june-2026-update>.
- [9] Aoun E. Muhammad et al. 2025. CORTEX: composite overlay for risk tiering and exposure in operational AI systems. <https://arxiv.org/abs/2508.19281>.
- [10] Kevin Paeth et al. 2025. Lessons for editors of AI incidents from the AI incident database. In *Proc. AAAI Conf. on Artificial Intelligence*. Vol. 39, 28946–28953. doi:10.1609/aaai.v39i2.8.35163.
- [11] Peter Slattery et al. 2026. The AI risk repository: a meta-review, database, and taxonomy of risks from artificial intelligence. *Patterns*, 101517. doi:10.1016/j.patter.2026.101517.